

Math 150 – Exam 3 – Chapter 4 – Fall 2015

I pledge to demonstrate personal and academic integrity in all matters with my fellow students, the faculty, and the administration here at USD. I promise to be honest and accountable for my actions; and to uphold the stature of scholastic honesty to better myself and those around me. Initials: _____

Name: _____ Section: **1 2**

Instructions: **Read all directions carefully. Ask for clarification if you do not understand the directions. Show your work when necessary. Complete sentences and justifications are only necessary when indicated.** There are 6 pages to this exam.

There are 100 points on this test and 4 Extra Credit points.

Num.	Score	Out of
1		3
2		15
3		12
4		20
5		20
6		20
7		10
B		*4*
Total		100

1. {3 points} Take a deep breath. Smile for 4 seconds.

2. {3 points each} State whether each statement is True or False.
 - (a) Let $a < b$. Suppose f is continuous on the closed interval $[a, b]$, differentiable on the open interval (a, b) , and $f(a) = f(b)$. Then there is a c in (a, b) with $f'(c) = 0$.

 - (b) If $f'(x) = 0$ when $x = 3$, then $f(3)$ is either a local minimum or a local maximum.

 - (c) If $f(a)$ is an absolute maximum and $f'(a) = 0$, then $f(a)$ is a local maximum.

 - (d) If $f'(x) = g'(x)$ for all real numbers x , then $f(x) = g(x)$.

 - (e) If $\lim_{x \rightarrow a} f(x) \cdot g(x)$ is of the indeterminate form $\infty \cdot 0$, then

$$\lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} f'(x) \cdot g'(x).$$

3. {12 points} Fill in the blanks:

- (a) If $f'(x) > 0$ for $-2 < x < 5$, then f is _____ on the interval $(-2, 5)$.

- (b) If $f''(2) < 0$ and $f'(2) = 0$, then f is concave _____ when $x = 2$ and $f(2)$ is a _____ of f .

- (c) If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is in indeterminate form of type _____
 and by _____ rule, $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} =$ _____ .

4. Let's get off to a good start. There are no wrong answers here. Write a complete sentence describing your greatest mathematical strength.

5. Evaluate each limit.

(a) $\lim_{x \rightarrow 2^-} \frac{1}{x-2} =$

(b) $\lim_{x \rightarrow 2^+} \frac{1}{x-2} =$

(c) $\lim_{x \rightarrow 2} \frac{1}{x-2} =$

(d) $\lim_{x \rightarrow 1} \frac{3x^4 - 2x^3 + 6}{x^2 + 3} =$

(e) $\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1} =$

(f) $\lim_{x \rightarrow \infty} \sqrt{x^2 - 1} - x =$

6. Let f be a function. Suppose $f'(a)$ is positive. Circle all the true statements.

- (a) The slope of f is increasing at a .
- (b) The function f is increasing at a .
- (c) The function f is concave up at a .
- (d) The function is continuous at a .

7. Suppose $f(x)$ is continuous at $x = 2$, $f(2) = 9$, and $\lim_{x \rightarrow 4} g(x) = 2$.

Compute $\lim_{x \rightarrow 4} [3f(g(x)) - x^2] =$

8. Use the Intermediate Value Theorem to determine if there is a root of $2x^3 + x^2 = -2$ in the interval $(-2, -1)$.

9. Recall the following two (equivalent) definitions of the derivative of a function f at a number $x = a$, denoted by $f'(a)$, are

Definition 1

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Definition 2

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

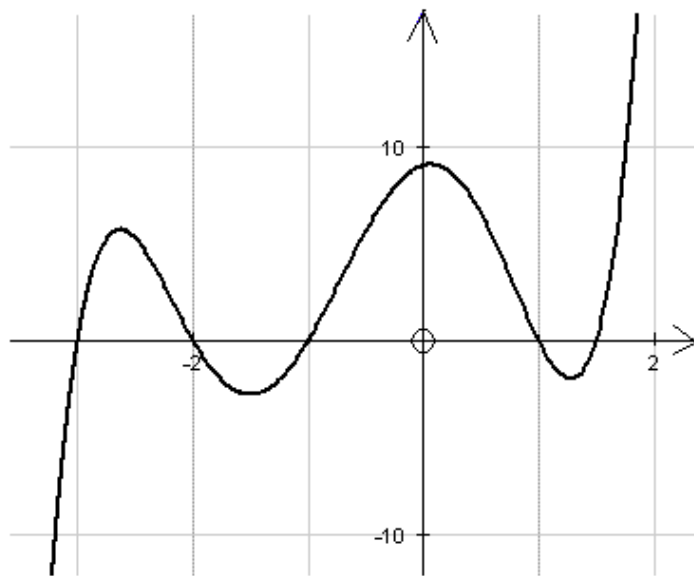
if this limit exists.

- (a) Given $f(x) = -2x^2 + 10$ and $x = 1$. Use either definition 1 or 2 to find $f'(x)$. You must show all your steps, but no explanation of steps are necessary.

- (b) Write a complete sentence that describes the geometric meaning of $f'(1)$.

10. List at least one vertical asymptote and at least one horizontal asymptotes (there are two of each type) of the following function $f(x)$ and use the definition of the type of asymptote to state why there is an asymptote:

11. Below is the graph of the velocity function of a particle. Sketch the graph of a position function of the particle.



12. Let

$$f(x) = \begin{cases} x^2 \cos(\frac{4}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Is $f(x)$ continuous at $x = 0$?

10 **Bonus Question: This is for extra credit**

Let if $f(x) = \sqrt{x} - x^2$. Apply Definition 1 from Question 6 to find the derivative $f'(x)$.

13. {4 points each} Evaluate the limits. Use l'Hopital's rule if necessary and possible. Be sure to show all your work.

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x - 2}$

(b) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$

(c) $\lim_{x \rightarrow \pi/2^+} \sec x - \tan x$

(d) $\lim_{x \rightarrow 0^+} x \ln x$

(e) $\lim_{x \rightarrow \infty} x e^{-x^2/2}$

14. Consider the function $f(x) = xe^{-x^2/2}$. The first derivative of f is $f'(x) = (1 - x^2)e^{-x^2/2}$ and the second derivative of f is $f''(x) = x(x - \sqrt{3})(x + \sqrt{3})e^{-x^2/2}$.

[Additional Hints]: $\sqrt{3} \approx 1.7$, and $f(\sqrt{3}) \approx 0.4$, $\lim_{x \rightarrow -\infty} xe^{-x^2/2} = 0$ and f is an odd function.

- (a) {2 points} What are the critical values of f .

- (b) {2 points} List the intervals in which f is increasing.

- (c) {2 points} List the intervals in which f is decreasing.

- (d) {2 points} List any local minimums and local maximums.

- (e) {2 points} List the intervals in which f is concave up.

- (f) {2 points} List the intervals in which f is concave down.

- (g) {2 points} Give any inflection points of f .

- (h) {2 points} State any vertical or horizontal asymptotes of f .

- (i) {4 points} Graph the function f on the graph below.

15. {4 points each} Find the most general antiderivatives.

(a) $f(x) = 5x^{1/4} - 7x^{3/4}$

(b) $f(x) = \frac{10}{x^9}$

(c) $f(x) = x(2 - x)^2$

(d) $f(x) = \frac{3}{x}$

(e) $f(x) = \sqrt[6]{x} - \sec^2(3x)$

16. {10 points} Recall, the area of a triangle is $A = (\text{base} \cdot \text{height})/2$. A right triangle is enlarged so that its base is growing at a rate of 2cm/s and its height is growing at a rate of 4 cm/s. How fast is the area growing when the base is 5cm and the height is 10cm?
{Be sure to show your work.}

*{4 points} * **Bonus Question: This is for extra credit**

Suppose $f'''(x) = \sin(x) + 2$, and $f''(0) = 1$, $f'(0) = 3$ and $f(0) = -5$.
What is $f(x)$?